THE INVERTIBLE MODULES ON A RINGED SPACE ARE REPRESENTED UP TO ISOMORPHISM BY A SET

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The local description of an invertible module as a direct summand of a finite free module. Invoked at the start of the counted proof (source lines 4540-4542) as a pointer to a lemma of this chapter. Bounds the local shape of the modules that have to be parametrised.
- The description of a sheaf by glueing data relative to an open covering. Invoked inside the counted proof (source lines 4550-4553) as a cross-chapter pointer. Turns the collection of locally described modules into a set of glueing data.

Source reference. Original lemma lemma-invertible: sources/stacks-modules/pinned-original-source.tex, source lines 4367–4367. Navigation pointer only: the counted text refers to the Stacks item lemma-invertible, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference sheaves-section-glueing-sheaves: sources/stacks-modules/pinned-original-source.tex, source lines 1199–1199. Navigation pointer only: the counted text refers to the Stacks item sheaves-section-glueing-sheaves, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. There exists a set of invertible modules $\{\mathcal{L}_i\}_{i \in I}$ such that each invertible module on X is isomorphic to exactly one of the $\mathcal{L}_i$.*

Proof. Recall that any invertible $\mathcal{O}_X$-module is locally a direct summand of a finite free $\mathcal{O}_X$-module, see Lemma lemma-invertible. For each open covering $\mathcal{U} : X = \bigcup_{j \in J} U_j$ and map $r : J \to \mathbf{N}$ consider the sheaves of $\mathcal{O}_X$-modules $\mathcal{F}$ such that $\mathcal{F}_j = \mathcal{F}|_{U_j}$ is a direct summand of $\mathcal{O}_{U_j}^{\oplus r(j)}$. The collection of isomorphism classes of $\mathcal{F}_j$ is a set, because $\mathrm{Hom}_{\mathcal{O}_U}(\mathcal{O}_U^{\oplus r}, \mathcal{O}_U^{\oplus r})$ is a set. The sheaf $\mathcal{F}$ is gotten by glueing $\mathcal{F}_j$, see Sheaves, Section sheaves-section-glueing-sheaves. Note that the collection of all glueing data forms a set. The collection of all coverings $\mathcal{U} : X = \bigcup_{j \in J} U_i$ where $J \to \mathcal{P}(X)$, $j \mapsto U_j$ is injective forms a set as well. For each covering there is a set of maps $r : J \to \mathbf{N}$. Hence the collection of all $\mathcal{F}$ forms a set. $\square$